

Supplementary Material: Proofs for “Spectral Costs of Multinomial and Residual Resampling in High-Dimensional Particle Ensembles: A Free-Probability Analysis”

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This supplement contains complete proofs of all results stated in the main text, in the order of the shared theorem counter used there. Section numbers S.0–S.22 are used throughout the main text’s proof citations.

S.0. TOOL INVENTORY

Tool S.1 (Rank inequality; [1, Lemma 2.2]). For Hermitian $d \times d$ matrices A, B , $\|F^A - F^B\|_\infty \leq \text{rank}(A - B)/d$.

Tool S.2 (Trace convexity chain; [2]–[6]). Ky Fan majorization; Karamata; Davis: φ convex $\Rightarrow M \mapsto \text{tr} \varphi(M)$ convex.

Tool S.3 (Extreme eigenvalues of sample covariance matrices; [7]). Let $Z \in \mathbb{R}^{d \times M}$ have i.i.d. entries drawn from a fixed distribution with mean 0, variance 1, finite fourth moment, and $d/M \rightarrow y \in (0, \infty)$. Then the extreme eigenvalues of $\frac{1}{M} Z Z^\top$ converge in probability to the edges of $\text{MP}(y)$: $\lambda_{\max} \rightarrow (1 + \sqrt{y})^2$; for $y < 1$, $\lambda_{\min} \rightarrow (1 - \sqrt{y})^2$. For $y > 1$, the Gram form gives directly: $\lambda_{\min}(\frac{1}{M} Z^\top Z) \rightarrow (\sqrt{y} - 1)^2 > 0$ in probability—implying simultaneously that Z has full column rank with probability tending to one and that the smallest positive eigenvalue of the sample covariance converges.

S.1. PROOF OF LEMMA 1

$$\mathbb{E}[S_{r,N} \mid X_N, w] = \frac{1}{N} \sum_i \mathbb{E}[n_i \mid X_N, w] x_i x_i^\top = \sum_i w_i x_i x_i^\top = S_{w,N}, \text{ by linearity only. } \square$$

S.2. PROOF OF THEOREM 3

The count vector of any scheme takes finitely many values $\mathbf{n}^{(1)}, \dots, \mathbf{n}^{(L)}$ with probabilities p_ℓ . By Lemma 1, $\sum_\ell p_\ell S(\mathbf{n}^{(\ell)}) = S_{w,N}$, and ordinary finite Jensen gives

$$\mathbb{E}[F(S_{r,N}) \mid X_N, w] = \sum_\ell p_\ell F(S(\mathbf{n}^{(\ell)})) \geq F(S_{w,N}).$$

For strictly convex F , equality forces $S(\mathbf{n}^{(\ell)})$ identical across the support, i.e. $S_{r,N} = S_{w,N}$ a.s.

Corollary (a) Tool S.2(the Davis trace-convexity step); for $q > 2$, $x \mapsto x^q$ is not operator convex on $(0, \infty)$, whereas trace convexity still applies. (b) $\lambda_{\max}(M) = \sup_{\|v\|=1} v^\top M v$, a supremum of linear functions. (c) $-\log(x + \varepsilon)$ is convex on $[0, \infty)$; the conclusion reads $\mathbb{E} \log \det(S_{r,N} + \varepsilon I) \leq \log \det(S_{w,N} + \varepsilon I)$, with the convention $0 \log 0 = 0$ throughout. $\square$

S.3. PROOF OF PROPOSITION 4 AND PROPOSITION 6

Proposition 4. With $\delta_i = n_i/N - w_i$, $\Delta := S_{r,N} - S_{w,N} = \sum_i \delta_i x_i x_i^\top$:

$$\text{tr } S_{r,N}^2 - \text{tr } S_{w,N}^2 = 2 \text{tr}(S_{w,N} \Delta) + \text{tr } \Delta^2, \quad \text{tr } \Delta^2 = \|\Delta\|_F^2.$$

The linear term vanishes in conditional mean (Lemma 1), so

$$\mathbb{E}[\text{tr } S_{r,N}^2 - \text{tr } S_{w,N}^2 \mid X_N, w] = \mathbb{E}[\|\Delta\|_F^2 \mid X_N, w] = \sum_{i,j} \mathbb{E}[\delta_i \delta_j] G_{ij}^2 = \frac{1}{N^2} \text{tr}[(G \circ G) \text{Cov}(\mathbf{n} \mid X_N, w)],$$

nonnegative since $G \circ G$ and the covariance are PSD (Schur product theorem).

Multinomial specialization. With $\mathbb{E}[n_i^2] = N w_i(1 - w_i) + N^2 w_i^2$ and $\mathbb{E}[n_i n_j] = N(N - 1) w_i w_j$ ($i \neq j$), substitution gives $\frac{1}{N} [\sum_i w_i \|x_i\|^4 - \text{tr } S_{w,N}^2]$. For the equivalent geometric form, set $Y_i := x_i x_i^\top$: since $\sum_i w_i = 1$ and $\sum_i w_i Y_i = S_{w,N}$,

$$\sum_i w_i \|Y_i - S_{w,N}\|_F^2 = \sum_i w_i \|Y_i\|_F^2 - \left\| \sum_i w_i Y_i \right\|_F^2,$$

so the cost equals $\frac{1}{N} \sum_i w_i \|x_i x_i^\top - S_{w,N}\|_F^2$, which vanishes iff $x_i x_i^\top = S_{w,N}$ for every i with $w_i > 0$ (covering all-zero particles and the $\pm v$ collinear case; in the latter, $S_{r,N} = \frac{1}{N} (\sum_i n_i) v v^\top = S_{w,N}$ a.s. even with random counts). $\square$

Proposition 6. For $x_i = \Sigma^{1/2} z_i$ with kurtosis κ_4 :

$$\mathbb{E}\|x_i\|^4 = (\text{tr } \Sigma)^2 + 2 \text{tr } \Sigma^2 + \kappa_4 \sum_j \Sigma_{jj}^2, \quad \mathbb{E}G_{ij}^2 = \text{tr } \Sigma^2 \quad (i \neq j).$$

Substituting into the multinomial formula and taking the expectation over the particle matrix (weights deterministic or independent):

$$\mathbb{E}[\text{tr } S_{r,N}^2 - \text{tr } S_{w,N}^2 | w] = \frac{1 - s_{2,N}}{N} \left[(\text{tr } \Sigma_N)^2 + \text{tr } \Sigma_N^2 + \kappa_4 \sum_j (\Sigma_N)_{jj}^2 \right],$$

exactly. For the asymptotics, $\frac{1}{d} \text{tr } \Sigma_N^2 \leq \|\Sigma_N\|_{\text{op}} \frac{\text{tr } \Sigma_N}{d} = O(1)$ (equivalently $\leq \|\Sigma_N\|_{\text{op}}^2$) and $\frac{1}{d} \sum_j (\Sigma_N)_{jj}^2 \leq \|\Sigma_N\|_{\text{op}}^2 = O(1)$, both under (A5) at finite N (no limit quantities); hence

$$\frac{1}{d} \mathbb{E}[\text{cost} | w] = c_N \left(\frac{\text{tr } \Sigma_N}{d} \right)^2 (1 - s_{2,N}) + O(N^{-1}) = c a_1^2 (1 - s_{2,N}) + o(1),$$

where no convergence rate is assumed for c_N or the population first moment, so the total error is only $o(1)$. $\square$

Corollary 25 (finite-sample Frobenius ordering). Write $\lambda = k + u$ with $L = \sum_i k_i$, $R = \sum_i u_i$, $L, R > 0$, and note $L + R = N$. In these coordinates $\text{Cov}_{\text{multi}} = \text{diag}(\lambda) - \frac{1}{N} \lambda \lambda^\top$ and $\text{Cov}_{\text{res}} = \text{diag}(u) - \frac{1}{R} u u^\top$, so, using $\text{diag}(\lambda) - \text{diag}(u) = \text{diag}(k)$,

$$\text{Cov}_{\text{multi}} - \text{Cov}_{\text{res}} = \text{diag}(k) - \frac{1}{N} \lambda \lambda^\top + \frac{1}{R} u u^\top.$$

Expanding the claimed rank-one term with LR/N and using $N = L + R$,

$$\frac{LR}{N} \left(\frac{k}{L} - \frac{u}{R} \right) \left(\frac{k}{L} - \frac{u}{R} \right)^\top = \frac{R}{LN} k k^\top - \frac{1}{N} (k u^\top + u k^\top) + \frac{L}{RN} u u^\top,$$

and the $k k^\top$, $k u^\top + u k^\top$, $u u^\top$ coefficients of $\text{diag}(k) - \frac{1}{N} \lambda \lambda^\top + \frac{1}{R} u u^\top$ minus $\text{diag}(k)$ are respectively $-\frac{1}{N}$, $-\frac{1}{N}$, $-\frac{1}{N} + \frac{1}{R}$, while those of $-\frac{1}{L} k k^\top + \frac{LR}{N}(\dots)$ are $-\frac{1}{L} + \frac{R}{LN} = -\frac{1}{N}$, $-\frac{1}{N}$, $\frac{L}{RN} = \frac{1}{R} - \frac{1}{N}$; the two sides agree, giving the stated identity. The first term $\text{diag}(k) - k k^\top / L$ is the covariance of a Multinomial($1; k/L$) draw, hence positive semidefinite, and the second term is an explicit positive semidefinite rank-one matrix, so the difference is positive semidefinite. The boundary cases are immediate: if $R = 0$ then $\text{Cov}_{\text{res}} = 0$; if $L = 0$ then $N = R$ and the two covariances coincide. The ordering of the conditional expected error energies follows from the quadratic form of Proposition 4, since $G \circ G \succeq 0$ (Schur product theorem) and $A \succeq B \succeq 0$ implies $\text{tr}[MA] \geq \text{tr}[MB]$ for $M \succeq 0$. $\square$

S.4. PROOF OF THEOREM 7 (MULTINOMIAL COUNT LAW)

Bounded test functions. Represent the counts by independent ancestor indices $J_1, \dots, J_N$ with $\mathbb{P}(J_a = i) = w_i$. Step 1 (expectation): the total-variation bound for the Poisson approximation is

$$d_{\text{TV}}(\text{Bin}(N, \lambda/N), \text{Poi}(\lambda)) \leq \frac{\lambda^2}{N},$$

which is $O(M^2/N)$ uniformly on $\lambda \in [0, M]$; the tail above M is controlled by (A4) via $\rho_N((M, \infty)) \leq M^{-2} \int \lambda^2 \rho_N \rightarrow M^{-2} r_2$. Hence $\frac{1}{N} \sum_i \mathbb{E} f(n_i) \rightarrow \int \mathbb{E} f(\text{Poi}(\lambda)) \rho(d\lambda)$. Step 2 (concentration): replacing one ancestor label changes two counts, so $\frac{1}{N} \sum_i f(n_i)$ changes by at most $4\|f\|_\infty/N$ (equivalently $2 \text{osc}(f)/N$); bounded differences give variance $O(N^{-1})$. (If weights are random, the computations above are conditional on w .)

Second moment (factorial-moment form). Let $B_N := \frac{1}{N} \sum_i n_i(n_i - 1)$, so $\frac{1}{N} \sum_i n_i^2 = 1 + B_N$. Conditionally on w ,

$$\mathbb{E}[B_N | w] = (N - 1) \sum_i w_i^2 = \left(1 - \frac{1}{N}\right) \int \lambda^2 \rho_N(d\lambda),$$

and expanding the collision indicators gives

$$\text{Var}(B_N | w) \leq 2 \sum_i w_i^2 + 4N \sum_i w_i^3 \rightarrow 0,$$

since $N \sum_i w_i^3 \leq (\max_i w_i) N \sum_i w_i^2 \rightarrow 0$ under (A4) (no third-moment assumption on the limiting weight law is needed). Hence $\frac{1}{N} \sum_i n_i^2 \rightarrow 1 + r_2 = b_2$ in probability. $\square$

S.5. POISSONIZATION COUPLING

Distributional identity. For independent $n'_i \sim \text{Poi}(\mu_i)$ with $R := \sum_i \mu_i$ a positive integer and $T = \sum_i n'_i$, for any k with $\sum k_i = R$,

$$\mathbb{P}(\mathbf{n}' = k \mid T = R) = \frac{\prod_i e^{-\mu_i} \mu_i^{k_i} / k_i!}{e^{-R} R^R / R!} = \frac{R!}{\prod_i k_i!} \prod_i \left(\frac{\mu_i}{R}\right)^{k_i} = \text{Multinomial}(R; \mu/R)\{k\}.$$

Coupling (label form). Let $T \sim \text{Poi}(R)$ independently of an infinite i.i.d. label sequence $J_1, J_2, \dots$ with $\mathbb{P}(J_a = i) = \mu_i/R$. Define n'_i as the number of occurrences of i among the first T labels, and n_i among the first R . Then (n'_i) are independent $\text{Poi}(\mu_i)$, $(n_i) \sim \text{Multinomial}(R; \mu/R)$, and $\sum_i |n_i - n'_i| = |T - R|$ exactly (the two count vectors differ only in the last $|T - R|$ labels). The matrix difference is a signed sum of $|T - R|$ rank-one terms, so $\text{rank}(S_N - S'_N) \leq |T - R|$; Tool S.1 gives $\|F^{S_N} - F^{S'_N}\|_\infty \leq |T - R|/d$, which tends to zero in probability provided $\sqrt{R}/d \rightarrow 0$ (satisfied here since $R \leq N$ and $d/N \rightarrow c > 0$). This coupling controls weak spectral convergence; it does not by itself control second spectral moments. $\square$

S.6. PROOF OF THEOREM 11 (LSD, IDENTITY POPULATION)

Step 1 (Poissonization). Coupling + Tool S.1: multinomial and independent-Poisson models share the same LSD limit; assume $D_N = \text{diag}(n_i)$ with independent $\text{Poi}(\lambda_i)$ entries.

Step 2 (truncation, in-probability two-limit scheme). Fix M and set $D_N^{(M)} := \text{diag}(n_i \wedge M)$. Then

$$\text{rank}\left(\frac{1}{N} Z(D_N - D_N^{(M)}) Z^\top\right) \leq \#\{i : n_i > M\}, \quad \frac{1}{N} \#\{i : n_i > M\} \xrightarrow{\mathbb{P}} \nu((M, \infty)),$$

so for every $\varepsilon > 0$,

$$\mathbb{P}\left(\|F^{S_{r,N}} - F^{S_{r,N}^{(M)}}\|_\infty > \frac{\nu((M, \infty))}{c} + \varepsilon\right) \rightarrow 0.$$

For each fixed M , Step 3 identifies the truncated model's limit as $\mathcal{T}_c(\nu^{(M)})$; the two-limit argument is then performed in Lévy or bounded-Lipschitz distance (not KS distance: the limit may have atoms), letting $M \rightarrow \infty$ with $\nu((M, \infty)) \rightarrow 0$ and continuity of the fixed-point law in the population law.

Step 3 (Silverstein convergence; random-diagonal transfer). With $W_N^{(M)} := \frac{1}{d}(D_N^{(M)})^{1/2} Z^\top Z (D_N^{(M)})^{1/2} \in \mathbb{R}^{N \times N}$: the empirical law of $D_N^{(M)}$ converges in probability, not per realization. From any subsequence, extract a further subsequence along which this convergence is almost sure; conditional on such a realization, apply the deterministic-population covariance theorem [1], [8]; bounded convergence of the conditional error probabilities then yields convergence in probability along the original subsequence. Hence $\mu_{W_N^{(M)}} \Rightarrow \mu_W^{(M)} = \nu^{(M)} \boxtimes \text{MP}(1/c)$ in probability.

Step 4 (exact companion bookkeeping). Let $Q_N^{(M)} := \frac{1}{N} Z D_N^{(M)} Z^\top \in \mathbb{R}^{d \times d}$ and $B_N^{(M)} := \frac{1}{N} (D_N^{(M)})^{1/2} Z^\top Z (D_N^{(M)})^{1/2} = \frac{d}{N} W_N^{(M)} \in \mathbb{R}^{N \times N}$. With $A = Z(D_N^{(M)})^{1/2}$, $B = (D_N^{(M)})^{1/2} Z^\top$, the products $AB = N Q_N^{(M)}$ and $BA = N B_N^{(M)}$ share nonzero spectra, and with $r = \text{rank}$, $c_N := d/N$:

$$\mu_{B_N^{(M)}} = c_N \mu_{Q_N^{(M)}} + (1 - c_N) \delta_0 \quad (\text{exact, every } N);$$

for $c_N > 1$ the negative coefficient is cancelled by the forced zero atom of $\mu_{Q_N^{(M)}}$.

Step 5 (assembly). $\mu_{B_N^{(M)}} \Rightarrow D_c(\mu_W^{(M)})$, so the truncated model converges to $\mathcal{T}_c(\nu^{(M)})$; Step 2's two-limit scheme yields $\mu_Q = \mathcal{T}_c(\nu)$. $\square$

Equivalence of the two forms of Definition 8. The companion transform of the W -model satisfies $\underline{m}_W(z) = c m_Q(cz)$ (from Step 4's scaling) and, by the classical companion form of Silverstein's equation with ratio $y = 1/c$,

$$z = -\frac{1}{\underline{m}_W(z)} + \frac{1}{c} \int \frac{t}{1 + t \underline{m}_W(z)} \nu(dt).$$

Substituting $\underline{m}_W(z) = c m_Q(cz)$, multiplying by c , and setting $\zeta = cz$ gives

$$\zeta = -\frac{1}{m_Q(\zeta)} + \int \frac{t}{1 + ct m_Q(\zeta)} \nu(dt),$$

the covariance-side fixed-point equation. $\square$

Normalization check. $\eta = \delta_1$: $B_N = \frac{1}{N} Z^\top Z$ has LSD $c \text{MP}(c) + (1 - c) \delta_0$ by Step 4's counting applied to the standard Wishart; independently, the covariance-side equation with $\nu = \delta_1$ reduces to the MP fixed point $czm^2 + (z + c - 1)m + 1 = 0$. Equating gives $\mathcal{T}_c(\delta_1) = \text{MP}(c)$ and $D_c(\text{MP}(1/c)) = c \text{MP}(c) + (1 - c) \delta_0$. (Moment recursion: $m_2 = 1 + cb_2$, and $m_3 = 1 + 3cb_2 + c^2b_3$ whenever the corresponding moments are finite.) $\square$

S.7. PROOF OF THEOREM 12 (GENERAL POPULATION SPECTRUM; GAUSSIAN CASE)

Order of argument: (i) truncate counts to $\nu^{(M)}$ as in S.6; the truncated $Q_N^{(M)}$ satisfies $\|Q_N^{(M)}\|_{\text{op}} \leq M \|\frac{1}{N} Z Z^\top\|_{\text{op}} = O_{\mathbb{P}}(1)$ (not a deterministic uniform bound over Gaussian realizations); (ii) given $D_N^{(M)}$ (independent of Z_N by exogeneity), Gaussianity makes $Q_N^{(M)}$ conditionally orthogonally invariant, hence asymptotically free from the deterministic Σ_N [9]; the probability modes are aligned as follows: the count-law convergence is in probability (S.6 Step 3's subsequence bridge), the freeness conclusion is applied conditionally, and the final LSD convergence is in probability; (iii) the truncated model's LSD is $H \boxtimes \mathcal{T}_c(\nu^{(M)})$; (iv) remove the truncation by Tool S.1 and pass through weak continuity of the free multiplicative convolution to $H \boxtimes \mathcal{T}_c(\nu)$. The matrices $\Sigma_N^{1/2} Q_N \Sigma_N^{1/2}$ and $Q_N^{1/2} \Sigma_N Q_N^{1/2}$ share their nonzero spectrum; for the bounded truncated model this relation, rather than any moment-sharing statement, is what the proof uses. $\square$

S.8. PROOF OF PROPOSITION 14 (ZERO-EIGENVALUE MASS)

Rank loss (finite sample). $S_{r,N} = \frac{1}{N} \sum_{i:n_i>0} n_i z_i z_i^\top$ has at most $\#\{i : n_i > 0\}$ nonzero rank-one summands, so $\text{rank}(S_{r,N}) \leq \min(d, \#\{i : n_i > 0\})$; equality a.s. for continuously distributed Z (general position). Note that a finite-sample rank proportion alone does not exclude a non-vanishing fraction of nonzero eigenvalues accumulating at zero (a vanishing fraction cannot form an atom, but a non-vanishing fraction tending to zero would matter), which is why the LSD statement is proved through the transform.

LSD zero mass (companion-law max-atom rule). With $q := \nu(\{0\})$, the max-atom rule for free multiplicative convolution [10], [11] applied to the companion law gives

$$\mu_B(\{0\}) = \max\{q, (1-c)_+\}, \quad \mu_B = D_c(\nu \boxtimes \text{MP}(1/c)),$$

since $\text{MP}(1/c)$ has zero mass $(1-c)_+$ (present iff $c < 1$). Reading off from $\mu_B = c\mu_Q + (1-c)\delta_0$:

$$\mu_Q(\{0\}) = \frac{\max\{q, (1-c)_+\} - (1-c)}{c} = \max\left(0, 1 - \frac{1-q}{c}\right).$$

For the separable case, $\mu_S(\{0\}) = \max\{H(\{0\}), \mu_Q(\{0\})\}$ is the same rule applied to $H \boxtimes \mu_Q$; the alternative rank route would require $\text{rank}(\Sigma_N)/d \rightarrow 1 - H(\{0\})$, which fails under (A5) alone (e.g. $\Sigma_N = N^{-1}I_d$), and is therefore not used. $\square$

S.9. PROOF OF PROPOSITION 15 (SPECTRAL MOMENTS; CONVERGENCE IN PROBABILITY)

Expectation identity. Let $Y := \Sigma^{1/2} Z D Z^\top \Sigma^{1/2}$ and $S = Y/N$; write everything as $\mathbb{E} \text{tr} Y^2 / (dN^2)$. Conditional Gaussian Wick evaluation given D :

$$\mathbb{E}[\text{tr} Y^2 \mid D] = ((\text{tr} \Sigma)^2 + \text{tr} \Sigma^2) \sum_i d_i^2 + \text{tr} \Sigma^2 \left(\sum_i d_i \right)^2.$$

For non-Gaussian entries an extra term $\kappa_4 \sum_i d_i^2 \sum_j \Sigma_{jj}^2$ appears, which after normalization by dN^2 is $O(N^{-1})$ under (A5) together with $\frac{1}{N} \sum_i d_i^2 = O(1)$ (in probability or in expectation, as recorded with each result).

Convergence in probability (auxiliary lemma). Under the standing assumptions, $\frac{1}{d} \text{tr} S_N \xrightarrow{\mathbb{P}} a_1$ and $\frac{1}{d} \text{tr} S_N^2 \xrightarrow{\mathbb{P}} m_{2,H} + c a_1^2 \cdot (\text{diag second moment})$.

First moment. Given D , $\mathbb{E}_Z[\frac{1}{d} \text{tr} S_N \mid D] = \frac{\text{tr} \Sigma_N}{d} \cdot \frac{\sum_i d_i}{N}$ and

$$\text{Var}_Z \left(\frac{1}{d} \text{tr} S_N \mid D \right) = \frac{2 \text{tr} \Sigma_N^2}{d^2 N^2} \sum_i d_i^2 \rightarrow 0,$$

since $\sum_i d_i = N$, $\frac{1}{N} \sum_i d_i^2 = O(1)$, and $\frac{1}{d} \text{tr} \Sigma_N^2 = O(1)$. Conditional Chebyshev plus convergence of the conditional mean give the claim for both pre and post matrices.

Second moment, two ingredients. (i) **Truncated model.** With $d_i^{(M)} := d_i \wedge M$,

$$\|S_N^{(M)}\|_{\text{op}} \leq M \|\Sigma_N\|_{\text{op}} \left\| \frac{1}{N} Z Z^\top \right\|_{\text{op}} = O_{\mathbb{P}}(1),$$

so the truncated model has asymptotically bounded spectrum, and its LSD convergence (to $H \boxtimes \mathcal{T}_c(\nu^{(M)})$, by S.7) upgrades to convergence in probability of its second spectral moment, whose limit is

$$m_{2,H} \left(\int t \nu^{(M)}(dt) \right)^2 + c a_1^2 \int t^2 \nu^{(M)}(dt)$$

(the truncated law has mean $\int t \nu^{(M)}(dt)$, generally less than one; the mean-one formula does not apply).

(ii) **Truncation-error control.** Set $X_{N,M} := \frac{1}{d} [\text{tr} S(D)^2 - \text{tr} S(D^{(M)})^2]$. Since $D \succeq D^{(M)} \succeq 0$, one has $S(D) \succeq S(D^{(M)}) \succeq 0$, and

$$\text{tr} A^2 - \text{tr} B^2 = 2 \text{tr} [B(A - B)] + \text{tr}(A - B)^2 \geq 0 \quad (A \succeq B \succeq 0),$$

so $X_{N,M} \geq 0$. With $\bar{d} := \frac{1}{N} \sum_i d_i$ and $\bar{d}_M := \frac{1}{N} \sum_i d_i^{(M)}$, the Wick identity applied to both matrices gives

$$\mathbb{E}_Z[X_{N,M} \mid D] = \frac{(\text{tr } \Sigma)^2 + \text{tr } \Sigma^2}{dN} \cdot \frac{1}{N} \sum_i (d_i^2 - (d_i^{(M)})^2) + \frac{\text{tr } \Sigma^2}{d} (\bar{d}^2 - \bar{d}_M^2),$$

and both terms vanish as $M \rightarrow \infty$ by the second tail moment of the count law (finite second moment; the mean difference $\bar{d} - \bar{d}_M \rightarrow 0$ likewise). To remove the truncation without assuming unconditional integrability of the random tail, let $A_{N,M}(D) := \mathbb{E}_Z[X_{N,M} \mid D]$; since $X_{N,M} \geq 0$, for any $\varepsilon, \delta > 0$,

$$\mathbb{P}(X_{N,M} > \varepsilon) \leq \mathbb{P}(A_{N,M} > \delta) + \frac{\delta}{\varepsilon},$$

because $\mathbb{P}(X > \varepsilon) \leq \mathbb{P}(A > \delta) + \mathbb{P}(X > \varepsilon, A \leq \delta)$ and the second term is at most $\mathbb{E}[A \mathbf{1}_{A \leq \delta}] / \varepsilon \leq \delta / \varepsilon$. First let $N \rightarrow \infty$ at fixed M , then $M \rightarrow \infty$, then $\delta \downarrow 0$; the truncation is removed.

The limit value follows from the expectation identity: pre-resampling ($D = \Lambda_N$): diagonal second moment $\rightarrow r_2$; post-resampling ($D = D_N$): $\rightarrow b_2 = 1 + r_2$ (Theorem 7). Increment $c a_1^2$. $\square$

S.10. PROOF OF PROPOSITION 16 (ESS; OCCUPANCY BOUND; THE PAIR)

ESS. $N/\text{ESS} = N \sum_i w_i^2 \rightarrow r_2$, equivalently $\text{ESS}/N \rightarrow 1/r_2$; given c and H , it determines the multinomial limiting second spectral moment (Proposition 15), but not the full limiting spectral distribution.

Occupancy bound (one line). For $K \sim \text{MixPoi}(\rho)$, $\mathbb{E}K = 1$ and $\mathbb{E}K^2 = 1 + r_2$; by Cauchy–Schwarz,

$$1 = (\mathbb{E}K)^2 = (\mathbb{E}[K \mathbf{1}_{\{K > 0\}}])^2 \leq \mathbb{E}K^2 \cdot \mathbb{P}(K > 0),$$

so $p_{\text{multi}} \geq 1/(1 + r_2)$. If $r_2 = 1$ with mean 1, then $\text{Var}(K) = 0$, forcing $\rho = \delta_1$ and $\nu = \text{Poi}(1)$. $\square$

The pair. $\rho^{(A)} = \frac{1}{2}\delta_{1/4} + \frac{1}{2}\delta_{7/4}$: mean 1; $\mathbb{E}L^2 = 25/16$; count-law zero mass $\frac{1}{2}e^{-1/4} + \frac{1}{2}e^{-7/4} = 0.476287$; count third moment $b_3 = 2.6875 + 3(25/16) + 1 = 8.375$. $\rho^{(B)} = \frac{9}{10}\delta_{3/4} + \frac{1}{10}\delta_{13/4}$: mean 1; $\mathbb{E}L^2 = 25/16$; zero mass $\frac{9}{10}e^{-3/4} + \frac{1}{10}e^{-13/4} = 0.429007$; $b_3 = 3.8125 + 3(25/16) + 1 = 9.5$. Identical ESS = $16N/25$ exactly at multiples of 10. Since $m_3(\mathcal{T}_c(\nu)) = 1 + 3cb_2 + c^2b_3$ (moments finite here) with equal $b_2 = 1 + r_2 = 41/16$ and different b_3 ,

$$m_3^{(B)} - m_3^{(A)} = c^2(9.5 - 8.375) = \frac{9}{8}c^2 > 0,$$

so the two limiting spectra differ for every $c > 0$ —this example is self-contained and does not rely on the general injectivity lemma (S.21). At $c = 1$, $\mu_Q(\{0\}) = \nu(\{0\})$ witnesses 0.476287 vs. 0.429007. (Differences in third spectral moments establish different distributions; they do not by themselves establish different support edges.) $\square$

S.11. PROOF OF COROLLARY 17 (EFFECTIVE DIMENSION; IPR COUNTEREXAMPLE)

Under the assumptions of Proposition 15, with $a_1 > 0$: S.9's convergence in probability of the normalized first and second moments, plus continuity of d_{eff}/d at the positive limit point, gives the two displayed limits and the explicit gap. No finite-sample ordering is claimed, because of the random denominator: with $d = N = 2$, $x_1 = 3e_1$, $x_2 = e_2$, $w = (0.9, 0.1)$, $\text{IPR}(S_w) = (8.1^2 + 0.1^2)/8.2^2 \approx 0.97591$, while multinomial resampling gives counts $(2, 0), (1, 1), (0, 2)$ with probabilities 0.81, 0.18, 0.01 and IPR values 1, 0.82, 1, hence the **conditional expectation** $\mathbb{E}[\text{IPR}(S_{r,N}) \mid X, w] = 0.82 + 0.18 \times 0.82 = 0.9676$ —strictly smaller. $\square$

S.12. PROOF OF THEOREM 18 (RANK-AND-SPECTRAL-GAP TRANSITION; CRITICAL TOUCHING)

Active-column reduction and domination. With $I_N := \{i : n_i > 0\}$,

$$S_{r,N} = \frac{1}{N} \sum_{i \in I_N} n_i z_i z_i^\top = \frac{1}{N} Z_{I_N} D_{\text{act}} Z_{I_N}^\top, \quad D_{\text{act}} \succeq I_{M_N},$$

since active counts are at least 1; hence $S_{r,N} \succeq \widetilde{W}_N := \frac{1}{N} Z_{I_N} Z_{I_N}^\top$ and Weyl monotonicity applies. Conditional on the active index set, Z_{I_N} has i.i.d. entries with the original entry distribution, by independence of the counts and the particle matrix (exogeneity). $M_N/N \rightarrow p$ in probability by Theorem 7. Random M_N is handled by conditioning and a subsequence argument (as in S.6 Step 3).

Subcritical $c < p$. $\widetilde{W}_N = \frac{M_N}{N} \cdot \frac{1}{M_N} Z_{I_N} Z_{I_N}^\top$; by Tool S.3, $\lambda_{\min}(\frac{1}{M_N} Z_{I_N} Z_{I_N}^\top) \rightarrow (1 - \sqrt{c/p})^2$ in probability, hence $\lambda_{\min}(\widetilde{W}_N) \rightarrow \frac{M_N}{N} (1 - \sqrt{c/p})^2 \rightarrow (\sqrt{p} - \sqrt{c})^2$ in probability, and Weyl monotonicity yields, for every $\varepsilon > 0$, $\mathbb{P}(\lambda_{\min}(S_{r,N}) \geq (\sqrt{p} - \sqrt{c})^2 - \varepsilon) \rightarrow 1$. Since the limit is positive, $\mathbb{P}(\text{rank}(S_{r,N}) = d) \rightarrow 1$ follows.

Supercritical $c > p$. Pass to the Gram matrix $\Gamma_N := \frac{1}{N} Z_{I_N}^\top Z_{I_N} = \frac{d}{N} \cdot \frac{1}{d} Z_{I_N}^\top Z_{I_N}$ ($M_N \times M_N$). The matrix $\frac{1}{d} Z_{I_N}^\top Z_{I_N}$ has population dimension M_N , sample size d , and ratio $M_N/d \rightarrow p/c < 1$ (note: $c > p$), so by Tool S.3 with $y = p/c < 1$, $\lambda_{\min}(\frac{1}{d} Z_{I_N}^\top Z_{I_N}) \rightarrow (1 - \sqrt{p/c})^2$ in probability; scaling by $d/N \rightarrow c$ gives $\lambda_{\min}(\Gamma_N) \rightarrow c(1 - \sqrt{p/c})^2 = (\sqrt{c} - \sqrt{p})^2$

in probability. The Gram lower-edge convergence also implies $\mathbb{P}(\text{rank}(Z_{I_N}) = M_N) \rightarrow 1$ (a positive limiting smallest eigenvalue forces full column rank with high probability; almost-sure rank equalities require continuous entries). On the event $\text{rank}(Z_{I_N}) = M_N$, both $\widetilde{W}_N$ and $S_{r,N}$ have rank exactly M_N (the latter by domination: $\text{range}(\widetilde{W}_N) \subseteq \text{range}(S_{r,N})$, and the active expansion gives the reverse inequality), and with the convention $\lambda_1 \geq \dots \geq \lambda_d$ (descending), the smallest positive eigenvalue is λ_{M_N} ; Weyl monotonicity at that index gives

$$\lambda_{\min}^+(S_{r,N}) = \lambda_{M_N}(S_{r,N}) \geq \lambda_{M_N}(\widetilde{W}_N) = \lambda_{\min}(\Gamma_N),$$

hence $\mathbb{P}(\lambda_{\min}^+(S_{r,N}) \geq (\sqrt{c} - \sqrt{p})^2 - \varepsilon) \rightarrow 1$. The zero-atom mass is Proposition 14.

Criticality $c = p$. The fixed-point identity of Definition 8, defined initially on $\mathbb{C}^+$, extends to the negative real axis by continuity (no singularity for nonnegative spectral measures). Write $p = \nu((0, \infty))$, set $c = p$, take $z = -s < 0$, and write $m = m(-s) > 0$. The equation rearranges to

$$sm(-s) = \frac{1}{c} \int_{(0, \infty)} \frac{\nu(dt)}{1 + ctm(-s)}.$$

Step 1 ($m(-s) \rightarrow \infty$). Suppose along some sequence $s_j \downarrow 0$ one had $m(-s_j) \leq C$. Then $\frac{1}{1+ctm(-s_j)} \geq \frac{1}{1+cC}$ pointwise, so the right side is bounded below by $\frac{1}{c} \int_{(0, \infty)} \frac{\nu(dt)}{1+cC} > 0$, while the left side $s_j m(-s_j) \rightarrow 0$ —contradiction. *Step 2 (no atom)*. Now that $m(-s) \rightarrow \infty$, the integrand tends to 0 pointwise and is bounded by 1; dominated convergence gives $sm(-s) \rightarrow 0$, and since $sm(-s) = \int \frac{s}{x+s} \mu_Q(dx) \rightarrow \mu_Q(\{0\})$, this proves $\mu_Q(\{0\}) = 0$. *Step 3 (touching)*. If the nonzero spectrum were separated from zero by a gap $\varepsilon > 0$, all mass would lie in $[\varepsilon, \infty)$ and $m(-s) = \int \frac{\mu_Q(dx)}{x+s} \leq 1/\varepsilon$, contradicting Step 1. Hence $\inf \text{supp } \mu_Q = 0$.

Finite-sample statement at criticality. For every $\varepsilon > 0$, $\mu_Q((0, \varepsilon)) > 0$ by Step 3; choose $f \in C_c((0, \varepsilon))$, $f \geq 0$, $\int f d\mu_Q > 0$. Empirical spectral convergence gives $\int f d\mu_N \xrightarrow{\mathbb{P}} \int f d\mu_Q > 0$, so with probability tending to one at least one eigenvalue lies in $(0, \varepsilon)$: $\mathbb{P}(\lambda_{\min}^+(S_{r,N}) < \varepsilon) \rightarrow 1$; nonnegativity gives the other direction, hence $\lambda_{\min}^+(S_{r,N}) \xrightarrow{\mathbb{P}} 0$. (No no-eigenvalues-outside-the-support input is needed for this statement; such inputs serve stronger claims, e.g. convergence to the true positive lower edge off criticality, or edge rates.)

Divergence rate of the Stieltjes transform. Multiplying the critical identity by $m(-s)$:

$$s m(-s)^2 = \frac{1}{c} \int_{(0, \infty)} \frac{m(-s)}{1 + ctm(-s)} \nu(dt).$$

As $s \downarrow 0$, $\frac{m(-s)}{1+ctm(-s)} \rightarrow \frac{1}{ct}$ pointwise and $\frac{m(-s)}{1+ctm(-s)} \leq \frac{1}{ct}$; for integer-valued counts, $t \geq 1$ on the support, so t^{-1} is integrable, and dominated convergence gives $s m(-s)^2 \rightarrow \frac{1}{c^2} \int_{(0, \infty)} \frac{1}{t} \nu(dt)$, i.e. $m(-s) \sim \frac{1}{c} \left(\int t^{-1} \nu(dt) \right)^{1/2} s^{-1/2}$. (This is an asymptotic for the Stieltjes transform along the negative real axis; a density-level $x^{-1/2}$ asymptotic would require additional argument.)

Benchmark scope. The domination argument discards the additional weights $D_{\text{act}} - I$, so it need not identify the exact lower edge. The exact regular lower edge depends on the full count law and is characterized by Proposition 19. $\square$

S.13. PROOF OF PROPOSITION 19 (EDGES; UNBOUNDED SUPPORT VIA POSITIVE FRACTIONS)

Edge parametrization. By [12] (see also [1, Ch. 6]), the law μ_W characterized by the Silverstein equation admits the real parametrization $z(u) = -\frac{1}{u} + y \int \frac{t}{1+tu} \nu(dt)$, $y = 1/c$, on real $u \neq 0$ with $-1/u \notin \text{supp } \nu$; the support-complement branches are those with $z'(u) > 0$, and regular endpoints arise at boundary critical points of these branches. Since $S_{r,N}$ and $B_N = \frac{d}{N} W_N$ share nonzero spectra and B_N is W_N scaled by $d/N \rightarrow c$, the regular edges of the nonzero part of μ_Q are $c z(u^*)$. Consistency: $\nu = \delta_1$ gives $c(1 \mp \sqrt{1/c})^2 = (1 \mp \sqrt{c})^2$. The critical zero endpoint corresponds to $u \rightarrow \infty$ rather than a finite critical point and is treated separately by Theorem 18(iii). $\square$

Unbounded support. Since ρ has mean 1, $\rho((0, \infty)) > 0$, so $\nu = \text{MixPoi}(\rho)$ satisfies $\nu(\{k\}) > 0$ for every $k \in \mathbb{Z}_{\geq 0}$; write $p_K := \nu([K, \infty)) > 0$, noting $p_K \rightarrow 0$. Fix a level $L > 0$ and choose K with $p_K < c/4$ and $Kc/4 > L$. Set $A_K := \{i : n_i \geq K\}$ and $B_{K,N} := \frac{K}{N} Z_{A_K} Z_{A_K}^\top$. Then $|A_K|/N \rightarrow p_K$ and, by Tool S.3 (and its Gram form, which gives full column rank of Z_{A_K} with probability tending to one), the smallest positive eigenvalue of $B_{K,N}$ converges in probability to

$$K(\sqrt{c} - \sqrt{p_K})^2 \geq K\left(\sqrt{c} - \frac{\sqrt{c}}{2}\right)^2 = \frac{Kc}{4} > L,$$

while with probability tending to one its nonzero eigenvalues number $|A_K|$, a fraction $|A_K|/d \rightarrow p_K/c > 0$ of the dimension. Since $S_{r,N} \succeq B_{K,N}$ (PSD order), Weyl monotonicity places at least $|A_K|$ eigenvalues of $S_{r,N}$ above $a_K := K(\sqrt{c} - \sqrt{p_K})^2$ with high probability. Now fix $L < \ell < a_K$ and take bounded continuous f with $f(x) = 0$ for $x \leq L$, $f(x) = 1$ for $x \geq \ell$, $0 \leq f \leq 1$. With probability tending to one,

$$\int f d\mu_N \geq \frac{|A_K|}{d},$$

so by weak convergence $\int f d\mu_Q \geq p_K/c > 0$; and since $f \leq \mathbf{1}_{(L,\infty)}$, $\mu_Q((L,\infty)) > 0$. As L was arbitrary, $\sup \text{supp } \mu_Q = \infty$; the same argument gives $\lambda_{\max}(S_{r,N}) \xrightarrow{\mathbb{P}} \infty$ in probability. Whether the support can split into several intervals for specific ρ is left open. $\square$

S.14. PROOF OF THEOREM 9 (RESIDUAL COUNT LAW; INCLUDING CONCENTRATION AND ALL BRANCHES)

Setup. $n_i = K_i + T_i$ with $K_i = \lfloor \lambda_i \rfloor$ deterministic and, when $R_N = \sum_i U_i > 0$, $\mathbf{T} \sim \text{Multinomial}(R_N; U/R_N)$; when $R_N = 0$, $\mathbf{T} = 0$. By (A4R), $\pi_N \Rightarrow \pi$ on $\mathbb{Z}_{\geq 0} \times [0, 1]$ and $R_N/N \rightarrow \mathbb{E}_\pi U$.

Weak convergence, case $\mathbb{E}_\pi U > 0$. Apply Lemma ?? with $(\mu_i) = (U_i)$, $R = R_N$: the residual counts couple to independent $\tilde{n}'_i \sim \text{Poi}(U_i)$ with rank defect $O(\sqrt{R_N}) = o(N)$; the diffuse condition on residual weights is automatic ($\max_i U_i/R_N \rightarrow 0$). For bounded $f : \mathbb{Z}_{\geq 0} \rightarrow \mathbb{R}$, set $g_f(k, u) := \mathbb{E}[f(k + \text{Poi}(u))]$, which is bounded and continuous on $\mathbb{Z}_{\geq 0} \times [0, 1]$; hence by (A4R), $\frac{1}{N} \sum_i g_f(K_i, U_i) \rightarrow \int g_f d\pi$ —this is the precise point at which the joint convergence $\pi_N \Rightarrow \pi$ (and not merely $\rho_N \Rightarrow \rho$) is needed. Concentration follows from the bounded-differences argument on the residual ancestor indices. Hence $\nu_N \Rightarrow \mathcal{L}(K + P_U)$.

Weak convergence, case $\mathbb{E}_\pi U = 0$. Then $R_N/N \rightarrow 0$, and

$$\#\{i : n_i \neq K_i\} \leq R_N = o(N),$$

so for bounded f , $|\frac{1}{N} \sum_i f(n_i) - \frac{1}{N} \sum_i f(K_i)| \leq 2\|f\|_\infty R_N/N \rightarrow 0$; by (A4R), $\pi_N \Rightarrow \pi$, and since $\mathbb{E}_\pi U = 0$ forces $U = 0$ π -a.s., $\mathcal{L}(K) = \mathcal{L}(K + U) = \rho$.

Second-moment convergence (all regimes). The following expectation and variance bounds hold for every $R_N > 0$, regardless of whether R_N/N tends to zero. Decompose

$$\frac{1}{N} \sum_i n_i^2 = \frac{1}{N} \sum_i K_i^2 + \frac{2}{N} \sum_i K_i T_i + \frac{1}{N} \sum_i T_i^2.$$

The exact expectation (for $R_N > 0$) is

$$\frac{1}{N} \sum_i \mathbb{E} n_i^2 = \frac{1}{N} \sum_i \lambda_i^2 + \frac{R_N}{N} - \frac{1}{NR_N} \sum_i U_i^2, \quad \left| \frac{1}{NR_N} \sum_i U_i^2 \right| \leq \frac{1}{N},$$

since $\mathbb{E} T_i^2 = U_i(1 - U_i/R_N) + U_i^2$ and $K_i^2 + 2K_i U_i + U_i^2 = \lambda_i^2$. For the variance: multinomial variances satisfy $\text{Var}(T_i) \leq U_i$ and cross-covariances are nonpositive, so with $q_i = U_i/R_N$,

$$\text{Var} \left(\frac{2}{N} \sum_i K_i T_i \right) \leq \frac{4}{N^2} \sum_i U_i K_i^2 \leq \frac{4}{N^2} \sum_i \lambda_i^2 = O(N^{-1}),$$

and the collision computation of S.4 with R_N ancestor draws and normalization N (using $\sum_i q_i^2 \leq 1/R_N$, $\sum_i q_i^3 \leq 1/R_N^2$) gives

$$\text{Var} \left(\frac{1}{N} \sum_i T_i^2 \right) \leq C \frac{R_N}{N^2} \leq \frac{C}{N}.$$

The two random terms are not independent, but independence is not needed: $\text{Var}(A + B) \leq 2\text{Var}(A) + 2\text{Var}(B)$. Hence $\frac{1}{N} \sum_i n_i^2 \rightarrow r_2 + \mathbb{E}_\pi U$ in probability in all regimes ($R_N = 0$ is deterministic and immediate). $\square$

S.15. PROOF OF THEOREM 20 (RESIDUAL SPECTRAL TRANSFORM)

ν_{res} has mean $\mathbb{E}[K + U] = 1$ and finite second moment $r_2 + \mathbb{E}_\pi U$. Silverstein's theorem transfers to a random diagonal D_N that is independent of Z_N with converging empirical law—it does not require the diagonal entries to be independent among themselves. Hence one may work with the raw residual count diagonal directly: (i) truncate the **total** counts, $D_N^{(M)} := \text{diag}(n_i \wedge M)$ (truncating only the Poisson part would not bound the diagonal, since K_i may be unbounded); (ii) by Theorem 9, the empirical law of $D_N^{(M)}$ converges to $\nu_{\text{res}}^{(M)}$; (iii) conditional on $D_N^{(M)}$, apply the covariance-spectrum theorem (with the subsequence bridge of S.6 Step 3); (iv) remove the truncation by the rank bound. This unifies the $\mathbb{E}_\pi U = 0$ and $\mathbb{E}_\pi U > 0$ cases; Poissonization is retained as a count-law tool only. (Two distinct truncation controls are used and should not be conflated: LSD rank truncation requires only the tail probability $\nu((M, \infty)) \rightarrow 0$; second-tail-moment control is needed for the empirical-moment arguments of S.9.) The Gaussian separable extension (S.7) is unchanged. $\square$

S.16. PROOF OF COROLLARY 21 (RESIDUAL ZERO ATOM AND THRESHOLD)

Under $\nu_{\text{res}} = \mathcal{L}(K + P_U)$, a zero count requires $K = 0$ and $P_U = 0$; on $\{K = 0\}$ the latter has probability e^{-U} (with $U = 1$ allowed in the limit, contributing e^{-1}). Hence $\nu_{\text{res}}(\{0\}) = \mathbb{E}_\pi[\mathbf{1}_{\{K=0\}} e^{-U}]$ and $p_{\text{res}}(\pi) = 1 - \nu_{\text{res}}(\{0\})$. Theorem 18's transfer: under the same identity-population, independence, and entry assumptions, the noncritical proof transfers through active-column domination, while the critical proof transfers through the residual LSD and its fixed-point equation. (For the occupancy step, $\mathbf{1}_{\{k>0\}}$ is continuous on the discrete count space; if the measures are placed on the real half-axis, replace it by a bounded continuous function agreeing with it at integers.) $\square$

S.17. PROOF OF PROPOSITION 22 (RESIDUAL SECOND-MOMENT COST)

The count-level identities $\mathbb{E}[K + P_U \mid K, U] = K + U$ and $\text{Var}(K + P_U \mid K, U) = U$ give $b_2(\nu_{\text{res}}) = \mathbb{E}[(K + U)^2] + \mathbb{E}[U] = r_2 + \mathbb{E}_\pi U$. The spectral-moment statement then follows from the residual version of S.9's argument (the truncation route of S.9 applies verbatim once the count second-moment convergence of S.14 is available, so the claim is an empirical-moment convergence, not merely a computation of the limit law's second moment). Strictness: $\mathbb{E}_\pi U \leq 1$ always (equality possible: the endpoint $\pi = \delta_{(0,1)}$), so the inequality is $\leq c a_1^2$, strict iff $a_1 > 0$ and $\mathbb{E}_\pi U < 1$; for $a_1 > 0$, equality with the multinomial cost occurs precisely at $\pi = \delta_{(0,1)}$, since $\mathbb{E}_\pi(K + U) = 1$ and $\mathbb{E}_\pi U = 1$ jointly force $U = 1$, $K = 0$ π -a.s. Vanishing: the cost vanishes iff $a_1 = 0$ or $\mathbb{E}_\pi U = 0$; assuming $a_1 > 0$, iff $\pi(\{U = 0\}) = 1$ —a property of the limit law, not of every finite sample (a vanishing fraction of exceptional weights is allowed). Exact finite- N zero cost: whenever counts are deterministic (in particular when every $\lambda_i \in \mathbb{Z}_{\geq 0}$, giving $R_N = 0$ and $n_i = \lambda_i$), or under the geometric degeneracy of S.3. $\square$

S.18. PROOF OF COROLLARY 23 (RANK-PRESERVATION ORDERING, TWO LEVELS)

The multinomial zero mass under the joint description: $\nu_{\text{multi}}(\{0\}) = \mathbb{E}_\pi[e^{-(K+U)}] = \mathbb{E}_\pi[e^{-L}]$. The residual zero mass (S.16): $\nu_{\text{res}}(\{0\}) = \mathbb{E}_\pi[\mathbf{1}_{\{K=0\}} e^{-U}]$. Difference:

$$\nu_{\text{multi}}(\{0\}) - \nu_{\text{res}}(\{0\}) = \mathbb{E}_\pi[\mathbf{1}_{\{K \geq 1\}} e^{-(K+U)}] \geq 0,$$

strict iff $\pi(K \geq 1) > 0$ (the integrand is strictly positive where $\mathbf{1}_{\{K \geq 1\}} = 1$). The spectral ordering $\mu_{\text{res}}(\{0\}) \leq \mu_{\text{multi}}(\{0\})$ follows since $q \mapsto \max(0, 1 - (1 - q)/c)$ is nondecreasing. For the spectral strictness, write $q_{\text{multi}} := (1 - p_{\text{multi}}/c)_+$ and $h_0 := H(\{0\})$; then $\mu_{\text{res}}(\{0\}) < \mu_{\text{multi}}(\{0\})$ iff $\pi(K \geq 1) > 0$ and $q_{\text{multi}} > h_0$ (in the identity-population case $h_0 = 0$, reducing to $c > p_{\text{multi}}$; for general H the inequality can be masked by a common atom h_0). In the window $c \in (p_{\text{multi}}, p_{\text{res}})$, multinomial resampling carries a macroscopic zero atom while residual resampling is of full rank with probability tending to one (identity population). $\square$

S.19. PROOF OF THEOREM 10 (CLASSIFICATION OF RESIDUAL LIMITS UNDER $\rho = \delta_1$)

Necessity. If $\rho_N \Rightarrow \delta_1$, then (A4R) forces π to be supported on $\{(k, u) : k + u = 1\}$, since the push-forward under the continuous map $(k, u) \mapsto k + u$ must be δ_1 . Since $k \in \mathbb{Z}_{\geq 0}$ and $u \in [0, 1]$, the line $k + u = 1$ meets the product set only at $(1, 0)$ and $(0, 1)$. Hence $\pi = (1 - \alpha)\delta_{(1,0)} + \alpha\delta_{(0,1)}$ for some $\alpha \in [0, 1]$, and Theorem 9 gives $\nu_{\text{res}} = (1 - \alpha)\delta_1 + \alpha \text{Poi}(1)$, with $\mathbb{E}_\pi U = \alpha$, $\nu_{\text{res}}(\{0\}) = \alpha e^{-1}$, $p_{\text{res}} = 1 - \alpha/e$, and second-moment cost $c a_1^2 \alpha$ (Proposition 22). $\square$

Realizability. (i) $\alpha = 0$: exactly uniform weights ($K_i = 1$, $U_i = 0$, $R_N = 0$). (ii) $\alpha = 1$ (bounded two-level endpoint construction): let $\lambda_i = 1 - \frac{1}{N}$ for $i < N$ and $\lambda_N = 2 - \frac{1}{N}$. Then $\sum_i \lambda_i = N$ exactly, $\max_i \lambda_i < 2$, $\frac{1}{N} \sum_i \lambda_i^2 = 1 + \frac{1}{N} - \frac{1}{N^2} \rightarrow 1$, and $\rho_N \Rightarrow \delta_1$; the first $N - 1$ particles have $(\lfloor \lambda_i \rfloor, \text{frac}(\lambda_i)) = (0, 1 - \frac{1}{N}) \rightarrow (0, 1)$, and the last has proportion $1/N \rightarrow 0$, so $\pi_N \Rightarrow \delta_{(0,1)}$; Theorem 9 then yields $\nu_{\text{res}} = \text{Poi}(1)$ directly, with no informal per-particle argument. (iii) $0 < \alpha < 1$: let $\alpha_N \rightarrow \alpha$ with $\alpha_N N$ integral and $\varepsilon_N \downarrow 0$; put a fraction α_N of particles at $1 - \varepsilon_N$ and the rest at $1 + \frac{\alpha_N}{1 - \alpha_N} \varepsilon_N$. The mean is exactly one: $\alpha_N(1 - \varepsilon_N) + (1 - \alpha_N)(1 + \frac{\alpha_N}{1 - \alpha_N} \varepsilon_N) = 1$; the exact second moment is

$$\frac{1}{N} \sum_i \lambda_i^2 = 1 + \frac{\alpha_N}{1 - \alpha_N} \varepsilon_N^2,$$

and the exact residual fraction is $R_N/N = \alpha_N$ (both used in experiment E4); the joint limit is $(1 - \alpha)\delta_{(1,0)} + \alpha\delta_{(0,1)}$. All constructions satisfy both (A4) and (A4R). $\square$

S.20. PROOF OF COROLLARY 24 (EXACT-UNIFORM-WEIGHT CONTRAST)

For $w_i = 1/N$ exactly: residual resampling leaves the second-moment matrix unchanged at every finite N ($n_i \equiv 1$ deterministically); in the identity-population setting its LSD is therefore $\text{MP}(c)$ (Theorem 20 with $\eta = \delta_1$, normalization check of S.6), with zero cost and $p_{\text{res}} = 1$; in the general Gaussian population model the unchanged matrix has LSD $H \boxtimes \text{MP}(c)$. Multinomial resampling gives $\text{Poi}(1)$, the distorted $\mathcal{T}_c(\text{Poi}(1))$, cost $c a_1^2$, and $p_{\text{multi}} = 1 - e^{-1}$. Both sides have $r_2 = 1$. The interpolation clause is Theorem 10. $\square$

S.21. LEMMA (INJECTIVITY OF $\mathcal{T}_c$; SELF-CONTAINED, ALL NONNEGATIVE LAWS)

Statement. For every $c > 0$, $\mathcal{T}_c$ is injective on the class of **all** probability laws on $[0, \infty)$.

Proof. Suppose $\mathcal{T}_c(\nu) = \mathcal{T}_c(\nu')$ and let m be the common Stieltjes transform. For every z in the upper half-plane, Definition 8's fixed-point equation gives

$$\int \frac{t}{1 + ctm(z)} \nu(dt) = \int \frac{t}{1 + ctm(z)} \nu'(dt).$$

Take $z = -s < 0$ and set $r := cm(-s) > 0$. The map $s \mapsto cm(-s)$ is continuous and strictly decreasing on any interval of the negative real axis ($\frac{d}{ds}m(-s) = -\int \frac{\mu(dx)}{(x+s)^2} < 0$), so as s varies over a nondegenerate interval, r varies over a nondegenerate interval $I \subset (0, \infty)$ —that is all the proof needs. Define

$$F_\nu(r) := \int \frac{t}{1+rt} \nu(dt) = \frac{1}{r} - \frac{1}{r^2} m_\nu\left(-\frac{1}{r}\right),$$

the identity holding since $\frac{1}{r} \int \frac{\nu(dx)}{1+rx} = \frac{1}{r} - \frac{1}{r^2} m_\nu(-1/r)$ by direct algebra. Equality of F_ν and $F_{\nu'}$ on I gives equality of m_ν and $m_{\nu'}$ on the corresponding negative interval; the identity theorem for analytic functions extends the equality to the common domain of analyticity, and uniqueness of the Stieltjes transform gives $\nu = \nu'$. (This route needs no mean-one assumption, no weighted measure, and no separate recovery of the atom at zero; the earlier cancellation-by- S -transform argument is retained only as a remark.) $\square$

S.22. ADDITIONAL REFERENCES FOR THE SUPPLEMENT

[S1] J. W. Silverstein and S. Choi, “Analysis of the limiting spectral distribution of large dimensional random matrices,” *J. Multivariate Anal.*, vol. 54, no. 2, pp. 295–309, 1995 (main-text [12]; edge parametrization). [S2] S. Belinschi, “The atoms of the free multiplicative convolution of two probability distributions,” *Integral Equations and Operator Theory*, vol. 46, no. 4, pp. 377–386, 2003 (main-text [10]; max-atom rule used in S.S.8). [S3] Z. D. Bai and Y. Q. Yin, “Limit of the smallest eigenvalue of a large dimensional sample covariance matrix,” *Ann. Probab.*, vol. 21, no. 3, pp. 1275–1294, 1993 (main-text [7]; Tool S.3). [S4] H. Bercovici and D. Voiculescu, “Free convolution of measures with unbounded support,” *Indiana Univ. Math. J.*, vol. 42, no. 3, pp. 733–773, 1993 (main-text [11]).

S.23. SOLVER UNIT TESTS

All spectral predictions in the main text are computed from the fixed-point equation of Definition 8 by two independent implementations (covariance side and companion side, the latter via the auxiliary variable $u = cm$ and $e = u + (c - 1)/z$). The battery below is run before any experiment; each row reports the realized tolerance.

Check	Quantity	Realized	Criterion
Branch	$\text{Im } m(z)$ at $z = i, 3 + 2i, 0.05 + 0.5i$	0.610, 0.263, 0.919	> 0
Equation residual	max over grid	2.2×10^{-13}	$\ll 10^{-8}$
Normalization	$z m(z)$ at $z = 10^6 i$	$-1.000000 + 10^{-6}i$	$\rightarrow -1$
MP(c) recovery	max interior relative density error	0.760%	$< 1\%$
Zero-mass consistency	max $ \text{solver} - \text{analytic} $, three configurations	4.3×10^{-8}	$\ll 10^{-6}$
Truncation $K=20$ vs $K=60$	max $ \Delta f $ on grid	2.1×10^{-15}	$\ll 10^{-8}$
	discarded second tail moment	5.1×10^{-15}	$\ll 10^{-8}$
Dual-solver cross-check	transform-relation residual	6.4×10^{-13}	$\ll 10^{-8}$
	companion identity $\int (1 + tu)^{-1} = -ze$ residual	1.9×10^{-13}	$\ll 10^{-8}$
	density deviation, $x > 0.15$	0.000%	$< 2\%$

The bandwidth sensitivity $\max |\Delta f| = 0.1438$ at $x = 0.020$ between $h = 10^{-3}$ and $h = 5 \times 10^{-4}$ is edge sharpening at the left boundary layer and is reported rather than tested; all conclusions in the main text are drawn away from this layer, and the $h/2$ sensitivity check reported there leaves every conclusion unchanged. The classical companion fixed-point forms found in the literature govern the companion transform of the W -side model and do not apply to the object studied here; the cross-check above is the reason the two implementations cannot silently agree on a wrong equation.

S.24. EXPERIMENT E2: FULL TWELVE-POINT SWEEP

Exactly uniform weights, so that $p_{\text{multi}} = 1 - e^{-1}$ and the threshold is $c = 1 - e^{-1} \approx 0.6321$. Part A is count-only (exact rank identity, $N = 10^4$, 300 repetitions); Part B reports the median and quartiles of $\lambda_{\min}^+$ over 15 repetitions at $N = 1000$, against the Marchenko–Pastur lower-edge benchmark $(\sqrt{c} - \sqrt{\bar{p}})^2$ of Theorem 18 and the true lower edge computed from the critical equation of Proposition 19.

c	$q_{0,N}$ empirical	q_0 theory	$ \text{gap} $
0.35	0	0	0
0.45	0	0	0
0.55	0	0	0
0.60	0	0	0
0.6321	0.00203 ± 0.00017	0	critical band
0.66	0.04223 ± 0.00027	0.04224	1×10^{-5}
0.72	0.12203 ± 0.00024	0.12205	3×10^{-5}
0.80	0.20997 ± 0.00023	0.20985	1×10^{-4}
0.90	0.29750 ± 0.00020	0.29764	1×10^{-4}
1.00	0.36803 ± 0.00019	0.36788	2×10^{-4}
1.08	0.41480 ± 0.00018	0.41470	1×10^{-4}
1.15	0.45029 ± 0.00016	0.45033	4×10^{-5}

The nonzero reading 0.00203 ± 0.00017 at the critical point $c = 0.6321$ is the finite-size critical-band smoothing annotated in the main text; at every off-critical grid point the empirical zero mass tracks $\max(0, 1 - (1 - e^{-1})/c)$ to within ± 0.001 .

c	$\lambda_{\min}^+$ median	quartiles	MP benchmark	true edge
0.35	0.05906	[0.05517, 0.06203]	0.04139	0.05607
0.45	0.02166	[0.02018, 0.02349]	0.01544	0.02058
0.55	0.00399	[0.00341, 0.00471]	0.00286	0.00376
0.60	0.00089	[0.00071, 0.00106]	0.00042	0.00055
0.6321	8.5×10^{-5}	$[2.4 \times 10^{-5}, 1.4 \times 10^{-4}]$	0	0
0.66	0.00037	[0.00019, 0.00054]	0.00030	0.00039
0.72	0.00359	[0.00316, 0.00432]	0.00286	0.00369
0.80	0.01433	[0.01313, 0.01499]	0.00987	0.01267
0.90	0.03050	[0.02884, 0.03126]	0.02360	0.03004
1.00	0.05428	[0.04954, 0.05957]	0.04200	0.05308
1.08	0.07949	[0.07602, 0.08137]	0.05962	0.07495
1.15	0.09721	[0.09564, 0.10371]	0.07691	0.09627

The median tracks the true edge at every grid point and collapses at criticality (8.5×10^{-5} at $c = p$), as Theorem 18(iii) requires. The true edge exceeds the benchmark at every off-critical point, by a factor between 1.25 and 1.27 over the upper half of the grid—the “approximately 1.3” conservativeness quoted in the main text.

S.25. EXPERIMENT E1: COUNT PROBABILITY MASS FUNCTIONS

Count probability mass functions for the design of Section VIII-B of the main text ($N = 1000$, $c = 0.8$), reported here at 200 paired repetitions as an independent higher-replication check of the same pre-registered design; the uniform–residual row is degenerate ($n_i \equiv 1$) and is omitted.

k	uniform, multinomial		two-point, multinomial		two-point, residual	
	theory	empirical	theory	empirical	theory	empirical
0	0.36788	0.36764 ± 0.00068	0.41483	0.41441 ± 0.00071	0.30327	0.30487 ± 0.00065
1	0.36788	0.36704 ± 0.00110	0.31898	0.31982 ± 0.00104	0.45490	0.45228 ± 0.00111
2	0.18394	0.18602 ± 0.00080	0.16342	0.16302 ± 0.00075	0.18954	0.19010 ± 0.00076
3	0.06131	0.06051 ± 0.00048	0.06907	0.06903 ± 0.00050	0.04423	0.04446 ± 0.00041
4	0.01533	0.01516 ± 0.00024	0.02432	0.02436 ± 0.00032	0.00711	0.00736 ± 0.00019
5	0.00307	0.00312 ± 0.00012	0.00714	0.00714 ± 0.00018	0.00087	0.00085 ± 0.00006

The theoretical columns are $\text{Poi}(1)$, $\text{MixPoi}(\frac{1}{2}\delta_{1/2} + \frac{1}{2}\delta_{3/2})$, and $\mathcal{L}(K + P_U)$ with (K, U) uniform on $\{(0, \frac{1}{2}), (1, \frac{1}{2})\}$ respectively, exactly as Theorems 7 and 9 predict; the zero row reproduces the zero masses of Table I of the main text within the respective error bars.

S.26. EXPERIMENT E4: FULL SWEEP

Full sweep $N = 2^8, \dots, 2^{14}$ for the three constructions of Theorem 10, extending Table III of the main text (which reports the endpoints). Columns: residual randomization fraction R_N/N (theory α), zero-count fraction (theory αe^{-1}), count second moment $b_{2,N}$ (theory $1 + \alpha$), and second-order weight deviation. For the uniform construction the counts are deterministic and all entries are exact.

Construction	N	R_N/N	zero fraction	$b_{2,N}$	$\frac{1}{N} \sum (\lambda_i - 1)^2$
interior ($\alpha = 0.4$)	256	0.3984	0.15408 ± 0.00089	1.3963 ± 0.0037	2.587×10^{-3}
	512	0.4004	0.1536 ± 0.0006	1.3997 ± 0.0027	1.304×10^{-3}
	1024	0.4004	0.1525 ± 0.0005	1.4046 ± 0.0020	6.521×10^{-4}
	2048	0.3999	0.1501 ± 0.0003	1.3993 ± 0.0014	3.254×10^{-4}
	4096	0.3999	0.1494 ± 0.0002	1.3997 ± 0.0010	1.627×10^{-4}
	8192	0.4000	0.1488 ± 0.0002	1.4000 ± 0.0007	8.139×10^{-5}
	16384	0.4000	0.1483 ± 0.0001	1.3998 ± 0.0005	4.069×10^{-5}
endpoint ($\alpha = 1$)	256	0.9961	0.36859 ± 0.00135	2.0034 ± 0.0062	3.891×10^{-3}
	512	0.9980	0.3681 ± 0.0010	1.9991 ± 0.0043	1.949×10^{-3}
	1024	0.9990	0.3669 ± 0.0007	1.9967 ± 0.0030	9.756×10^{-4}
	2048	0.9995	0.3680 ± 0.0005	2.0003 ± 0.0021	4.880×10^{-4}
	4096	0.9998	0.3678 ± 0.0003	1.9979 ± 0.0015	2.441×10^{-4}
	8192	0.9999	0.3675 ± 0.0002	1.9998 ± 0.0011	1.221×10^{-4}
	16384	0.9999	0.3680 ± 0.0002	2.0007 ± 0.0008	6.103×10^{-5}
uniform	all	0	0	1	0
theory		α	αe^{-1}	$1 + \alpha$	$\rightarrow 0$

The zero-count fraction converges to αe^{-1} from above at rate consistent with $N^{-1/2}$, and the second-order deviation decreases by a factor of 4 per doubling of N in all three constructions, while R_N/N remains at its theoretical limit throughout.

S.27. ROBUSTNESS CHECKS IN FULL

Scale checks (A1). E1 repeated at $N = 500$ and $N = 2000$ (same $c = 0.8$, 20 repetitions):

N	profile	m_2 (th)	$\nu(\{0\})$ (th)	$\mu_Q(\{0\})$ (th)
500	uniform	2.5860 ± 0.0210 (2.6)	0.3646 ± 0.0051 (0.36788)	0.20575 ± 0.00637 (0.20985)
500	two-point	2.8327 ± 0.0142 (2.8)	0.4154 ± 0.0042 (0.41483)	0.26925 ± 0.00525 (0.26854)
2000	uniform	2.6184 ± 0.0081 (2.6)	0.3701 ± 0.0020 (0.36788)	0.21256 ± 0.00244 (0.20985)
2000	two-point	2.7992 ± 0.0102 (2.8)	0.4145 ± 0.0020 (0.41483)	0.26806 ± 0.00247 (0.26854)

Every quantity converges toward its prediction with error bars shrinking at the Monte Carlo rate; deviations at $N = 2000$ are within 1.4σ everywhere.

Non-Gaussian entries (A2). Two-point profile, multinomial resampling, $c = 0.8$, variance-one entries with kurtosis κ_4 ; finite- N predictions from Proposition 6 include the kurtosis term:

entries	κ_4	m_2 empirical	m_2 finite- N theory	$\mu_Q(\{0\})$ empirical (th 0.26854)
Gaussian	0	2.800 ± 0.012	2.80000	0.2705 ± 0.0052
Rademacher	-2	2.821 ± 0.017	2.79800	0.2755 ± 0.0047
Student- t_8	1.5	2.821 ± 0.021	2.80150	0.2690 ± 0.0051

Deviations from the finite- N predictions are 1.3σ and 0.9σ . This is an agreement report for a single configuration; no universality claim is made.

Separable population spectrum (A3). $H = \frac{1}{2}\delta_{1/2} + \frac{1}{2}\delta_{3/2}$, uniform weights, multinomial resampling, $c = 0.8$: post-resampling $m_1 = 1.0011$ (theory $a_1 = 1$); m_2 pre 2.054 (theory 2.05) and post 2.856 (theory 2.85); zero atom 0.2079 (theory 0.20985 by the max-atom rule of Proposition 14). As an end-to-end control, the $H \boxtimes \mathcal{T}_c(\text{Poi}(1))$ pipeline (S-transform computation) reproduces a directly simulated separable Wishart spectrum with maximum Stieltjes deviation 5.1×10^{-4} (Fig. 1).

Two-point population spectrum

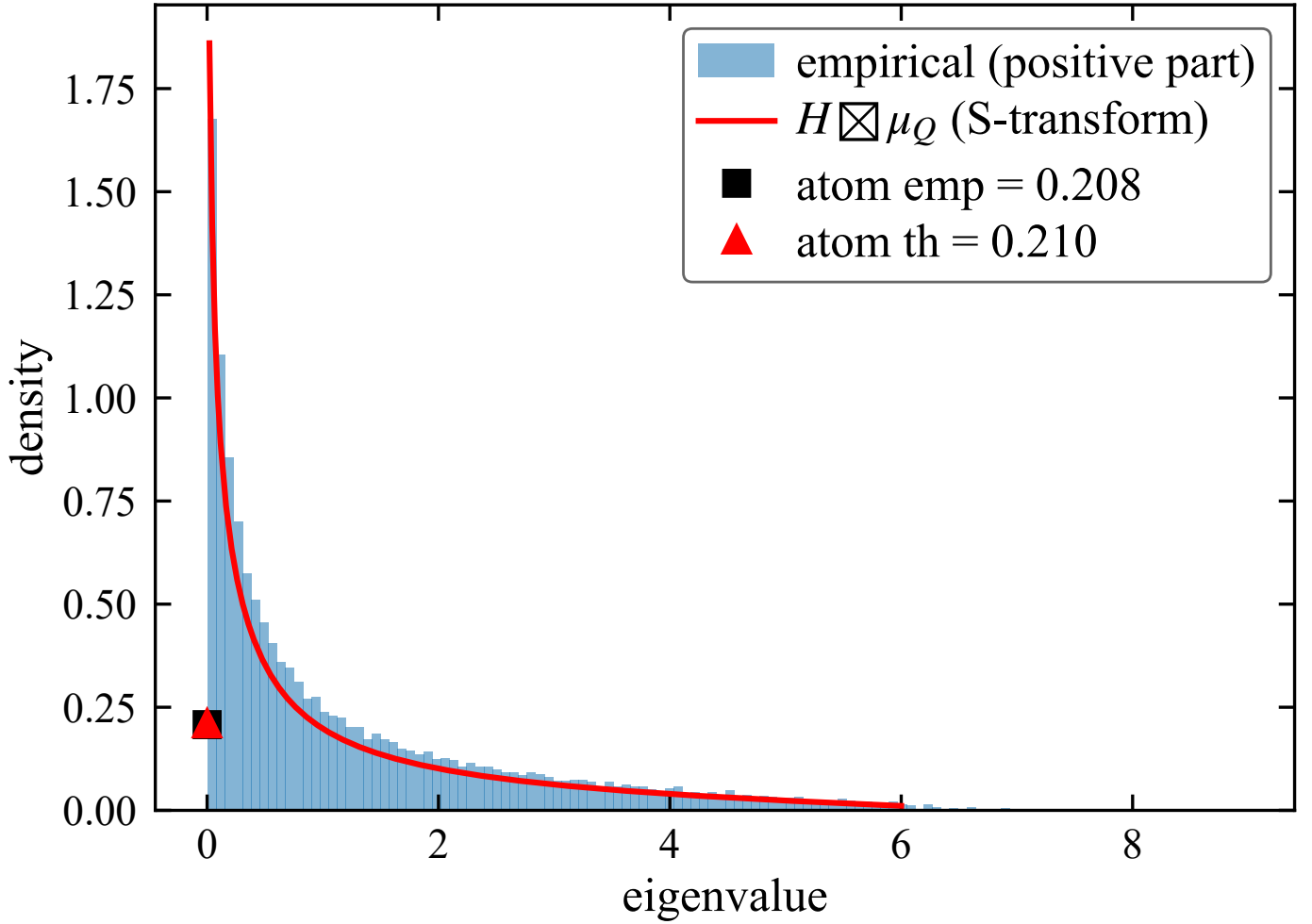


Fig. 1. End-to-end check of Theorem 12: directly simulated spectrum under the two-point population $H = \frac{1}{2}\delta_{1/2} + \frac{1}{2}\delta_{3/2}$ against the $H \boxtimes \mathcal{T}_c(\text{Poi}(1))$ prediction, with the zero atom drawn separately. The two computation routes agree to a maximum Stieltjes deviation of 5.1×10^{-4} .

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